AIOS Score Formulas

How every sprint CRI score and Benchmark Assessment score is calculated · 29 October 2026 (checked against the code)

Sprint CRI scores

Which facts count

Only **confirmed** or **corrected** facts are evidence. A fact is confirmed by a person, or automatically by the fact-verification rules when `FACT_VERIFICATION_MODE=enforce` (in the default *shadow* mode the rules only record what they would do). The AI's confidence never confirms a fact on its own. Extracted, rejected and evidence-requested facts never count.

1. Raw / Sprint CRI (0–100) — Live CRI Preview

The CRI has 9 factors (the 9 AI-QIF criteria). Each factor has one or more sub-criteria, each with a weight. **Current configuration:** each factor has one sub-criterion, *General evidence* (weight 1), which takes every confirmed or corrected fact of that factor. So today a factor's criteria score is simply the average AI confidence of all its confirmed / corrected facts × 100.

Step 1 — each sub-criterion's score

```
sub-criterion score = average AI confidence of its confirmed/corrected facts × 100
```

A sub-criterion with a field list takes only facts of those fields; one without (General evidence) takes every confirmed/corrected fact of the factor. It is 0 if no such fact exists.

Step 2 — each factor's score

```
criteria score = Σ(sub-criterion score × sub-criterion weight)
                 / Σ(sub-criterion weights)
gap penalty    = Σ penalty for each unresolved gap on the factor
                 (blocking 8, high 5, medium 3, optional 1)
factor score   = criteria score − gap penalty,
                 kept between 0 and 100, rounded to 2 decimals
```

Unresolved gaps are those whose status is **Open**, **In progress** or **Needs replacement** (a rejected value still waiting for a replacement). Resolved, skipped and unavailable gaps cost nothing.

Step 3 — the CRI

```
contribution = factor score × factor weight, rounded to 2 decimals
               (weights add up to 1, e.g. 0.12 = 120/1000)
Raw CRI      = Σ contributions of the 9 factors,
               kept between 0 and 100, rounded to 2 decimals
```

Rounding happens per factor and per contribution before the sum, so the CRI can differ by about 0.01 from a calculation without intermediate rounding. Weights are not re-normalised: an inactive factor contributes 0.

Factor status (shown in the breakdown, not part of any score), checked in this order:

- **Not started:** no confirmed / corrected fact.
- **At risk:** an unresolved blocking gap, or a score below 40.
- **Developing:** a score from 40 to 69.99.
- **Strong:** a score of 70 or more.

2. CRI Evidence Confidence (0–100%) — Extracted Facts Review

```
factor confidence = Σ(sub-criterion average confidence × sub-criterion weight)
                    / Σ(sub-criterion weights)
CRI confidence    = Σ(factor confidence × factor weight), kept between 0 and 100%
band              = High if ≥ 80%, otherwise Building
```

Gap penalties don't affect confidence.

3. AI-QIF Evidence Coverage (0–100%) — formula AIQIF-COV-1

Indicator weight. If the published benchmark release gives every indicator an approved weight, those weights are used. Otherwise:

```
indicator weight = factor weight / number of indicators in that factor
```

Applicable. Every indicator counts unless the release's rules say it does *not apply* to the college's benchmark profile. If applicability can't be decided (no published release, no benchmark profile, or the indicator is not in the release), the indicator still counts. "Optional" indicators count too.

Covered. At least one tag on that indicator counts. A tag counts when all of these hold:

- its fact is **AI evidence** (not a context or institution-profile fact) and is confirmed or corrected;
- the tag's relationship is **Direct** or **Supporting** (Contextual never counts);
- the tag was accepted — by a Super Admin, or automatically (the NAAC → AI-QIF crosswalk, a confident AI tag of the fact's own criterion, a matching earlier expert decision, or the AI second opinion) — and was never rejected.

Tags still waiting in the AI-QIF expert queue don't count; while the queue is not empty the coverage is marked provisional. Two facts on one indicator count it once.

```
coverage = Σ weight of covered applicable indicators
           / Σ weight of all applicable indicators
```

4. Coverage-Adjusted CRI (0–100) — Live CRI Preview

```
Coverage-Adjusted CRI = Raw CRI × coverage, rounded to 2 decimals
```

Example: 89 × 0.75 = 66.75. It shows "—" when no indicator is applicable.

Benchmark Assessment scores (per benchmark, each 0–100)

These are calculated separately and don't affect the sprint CRI.

Score	Formula
Compliance	100 × (mandatory foundational L3 evidence requests met / total mandatory foundational L3 evidence requests), rounded to 2 decimals. Requests without a level count as foundational. Blank for InGage-proposed benchmarks (no legal requirement), and blank when the release has no mandatory foundational request.
Maturity	Level reached: none = 0, L3 = 33.33, L2 = 66.67, L1 = 100. A level counts only when every mandatory request for it (and every mandatory request without a level) and the levels below it is met. An assessor can override the level.
Evidence confidence	The average review quality of the evidence that counts (accepted, current), using each piece's latest review. Each piece's quality = average of its 5 review scores (relevance, completeness, currency, authenticity, scope). 0 when no evidence counts.
Outcome	Entered by the assessor from measured results, with a note. Blank until entered.
Composite	(compliance × weight) + (maturity × weight) + (evidence × weight) + (outcome × weight), rounded to 2 decimals. The four weights must add up to 1 and be approved by the benchmark owner for the active scoring model. Blank while there are no approved weights, or while any part with a weight above 0 is blank.

A request is **met** when at least its minimum number of pieces of evidence count. A piece counts only if it is accepted, not expired on the cycle's opening date, and, where the rule requires it, issued within the required number of months before that date.

Source: backend/apps/scoring/services/cri_engine.py, backend/apps/scoring/services/coverage.py, backend/apps/facts/aiqif_tagging.py (counting_q), backend/apps/gaps/constants.py, backend/apps/assessments/services.py. Fact confirmation: backend/apps/facts/verification/.